Lesson 3.18 - Mean Absolute Deviation

> **Video:** Mean Absolute Deviation by The Organic Chemistry Tutor **Chapter:** 3 - Central Tendency

Idea

The **mean absolute deviation (MAD)** is the **average distance** of the data values from the mean.

$$\text{MAD} = \frac{\sum |x - \bar{x}|}{n}$$

Steps

1. Find the mean $\bar{x}$.
2. Find each deviation $x - \bar{x}$.
3. Take **absolute values**, so all distances are positive.
4. Add them and divide by n.

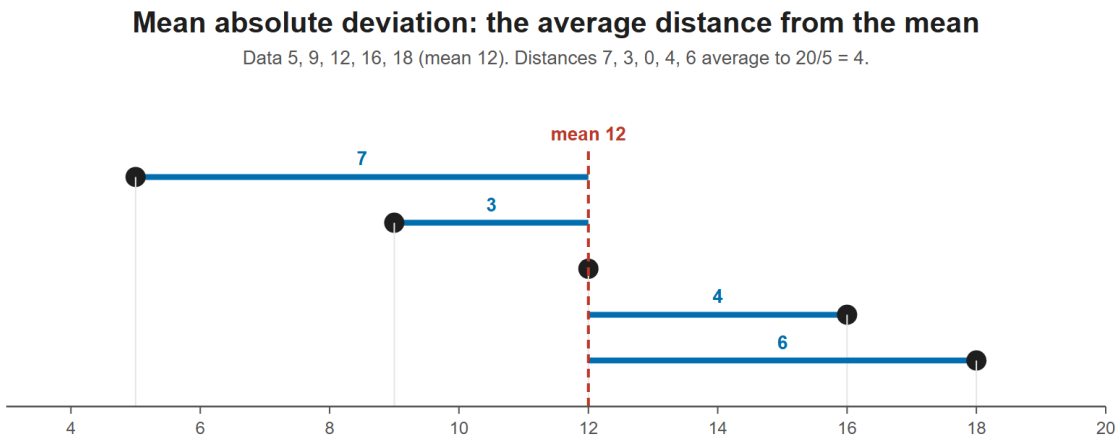
Example 1 - with a table

Data: 7, 11, 14, 19, 22, 29

Mean: $\bar{x} = \frac{\sum x}{n} = \frac{102}{6} = 17$

x	$x - \bar{x}$	$\lvert x - \bar{x} \rvert$
7	−10	10
11	−6	6
14	−3	3
19	2	2
22	5	5
29	12	12
Sum	0	**38**

$$\text{MAD} = \frac{\sum |x - \bar{x}|}{n} = \frac{38}{6} = 6\tfrac{1}{3} \approx 6.33$$



Each bar is the distance of one value from the mean of 12. The mean absolute deviation is the average bar length, (7 + 3 + 0 + 4 + 6)/5 = 4.

Example 2 - directly with the formula

Data: 5, 9, 12, 16, 18

Mean: $\bar{x} = \dfrac{\sum x}{n} = \dfrac{60}{5} = 12$

$$\text{MAD} = \frac{\sum |x - \bar{x}|}{n} = \frac{|5-12| + |9-12| + |12-12| + |16-12| + |18-12|}{5} = \frac{7+3+0+4+6}{5} = \frac{20}{5} = 4$$

What it means

Each absolute deviation is a distance from the mean 12 on the number line:

Value	5	9	12	16	18
Distance from 12	7	3	0	4	6

The MAD is the average of these five distances: $(7 + 3 + 0 + 4 + 6)/5 = 4$. On average, each value is **4 units** from the mean.

Key takeaways

- MAD = average of the **absolute** deviations from the mean.
- Absolute values are needed because the plain deviations always add up to **zero** (see Lesson 3.19).
- MAD is easier to interpret than the standard deviation (no squaring), but the SD is used more often in later statistics.